

From Leaderboard Ranks to Audited Decisions: A Controlled-Panel Study of Evaluation-Format Uncertainty

Anonymous ACL submission

Abstract

A persistent evaluation gap in model leaderboards is that rank tables often hide the prompt, scorer, metric, and sampling choices that determine whether one model can be responsibly described as better than another. Prior work makes benchmark and prompt sensitivity visible, but it leaves a practical author-facing question open: after a fixed set of official scorer outputs has been collected, which pairwise model claims should a paper state, soften, or withhold? We introduce a controlled-panel pairwise decision audit that converts lm-evaluation-harness outputs into stable-or-abstain reports: within each benchmark family and metric, a model pair receives a direction only when the chosen confidence-band rule supports it, and otherwise the paper records abstention. We evaluate the audit on 2560 scored rows for 10 open instruction/chat models across 5 benchmark families, with same-output baselines that approximate prompt marginalization, rank-interval uncertainty, and benchmark perturbation summaries. The main result is that simultaneous bands yield 15 stable decisions and 300 abstentions over 315 model-pair/family/metric comparisons, while naive intervals would report 43 stable decisions and the observed evaluator-format axis produces 53 direction-flipping pairs. These findings turn uncertainty into a reporting standard for the controlled panel: publish directional claims when the interval rule authorizes them, and otherwise state abstention and the missing evidence needed for stronger public-leaderboard claims.

1 Introduction

Public leaderboards compress benchmark choice, prompt or formatting convention, scorer behavior, and metric aggregation into a single ordering. That compression is useful for scanning models, but it creates an evaluation gap when a practitioner treats an adjacent rank as a reliable pairwise recommendation. In that setting, the scientific question is

not only where a model sits in an ordered list; it is whether a paper should turn a small score gap into a directional sentence about one model being better than another.

Prior work gives strong reasons to be cautious. Benchmark-targeting studies show that rankings can move under prompt and scoring perturbations (Alzahrani et al., 2024b,a), and efficient multi-prompt evaluation and rank-uncertainty methods give tools for aggregating or intervalizing such variation (Polo et al., 2024b; Neuhof and Benjamini, 2026; Avelar Menendez et al., 2026). Those methods make uncertainty visible, but a manuscript still needs an author-facing decision rule after its scorer tensor is fixed: state a direction, soften it, or withhold it.

We therefore treat leaderboard validity as a claim-reporting problem rather than as another rank-correlation metric. A rank table always orders systems, even when score differences are inside the noise induced by task sampling, evaluator settings, or metric choice; a reporting decision can instead say “we do not know.” Earlier NLP evaluation work has called for stronger reporting of variance, significance, and experimental conditions (Dror et al., 2018; Dodge et al., 2019; Card et al., 2020). Our audit makes that discipline operational: it starts from official scorer outputs, forms model pairs within each benchmark family and metric, applies naive and simultaneous confidence intervals, and labels each pair as stable or abstained. A separate evaluator-format audit checks whether the observed chat-template versus no-chat axis changes the pair direction, so sensitivity informs the claim without being mistaken for a certified public-leaderboard reversal.

The empirical question is whether this stable-or-abstain view changes what the paper should report. The completed run covers MMLU (Hendrycks et al., 2021), ARC-Challenge (Clark et al., 2018), HellaSwag (Zellers et al., 2019), GSM8K (Cobbe

et al., 2021), and TruthfulQA (Lin et al., 2022) through the lm-evaluation-harness scorer (LM Evaluation Harness Contributors, 2023). It includes ten open model identifiers, two observed evaluator-format axes, 2560 full scored rows, and 778 same-output baseline rows. We find that simultaneous bands yield only 15/315 reportable pairwise decisions, a 4.8% stable-decision rate, while naive intervals would report 43 stable decisions over the same pair set and the evaluator-format audit records 53 direction-flipping pairs. Abstention is therefore a measured output of the reporting rule, not an apology for failing to produce a rank.

We make three contributions. First, we define a pairwise decision audit that reports stable decisions and abstentions instead of treating every rank difference as actionable. Second, we compare that audit with same-output PromptEval-style, Rank-Intervals-style, and benchmark-perturbation summaries on the same tensor. Third, we provide a claim downgrade ledger that separates supported controlled-panel findings from claims that would require frozen public leaderboard rows, validated tinyBenchmarks/IRT subsets (Polo et al., 2024a), or the unexecuted shot, seed, option-order, and GSM8K extraction axes.

2 Related Work

Benchmark and leaderboard sensitivity. Benchmark-targeting work shows that model rankings can shift under prompt, answer-symbol, scoring, and benchmark perturbations (Alzahrani et al., 2024b,a; Razavi et al., 2025; Oyarhoseini et al., 2026). Related leaderboard critiques argue that benchmark incentives can diverge from user utility and that static benchmarks can be gamed or saturated (Ethayarajh and Jurafsky, 2020; Kiela et al., 2021; Contrast Sets Authors, 2020). This paper does not claim that generic sensitivity is new. It uses that prior result as motivation for a downstream reporting question: whether a pairwise conclusion should be stated, softened, or withheld.

Prompt marginalization and rank uncertainty. PromptEval-style marginalization estimates performance across prompt choices (Polo et al., 2024b), while rank intervals and related uncertainty methods quantify uncertainty in leaderboard positions (Neuhof and Benjamini, 2026; Heineman et al., 2025; Kargi and Salinas, 2026; Huang et al., 2026; Long, 2026). Our protocol is simpler and more

conservative: it asks whether a pairwise claim is stable under same-output intervals and the observed evaluator-format axis. This makes abstention a first-class result.

Statistical reporting and metric breadth. NLP has long treated automatic metrics and significance tests as reporting devices rather than guarantees of task utility (Papineni et al., 2002; Lin, 2004; Dror et al., 2018). Recent broad evaluation suites extend that lesson to LLMs by making scenario coverage, metric multiplicity, and benchmark incompleteness explicit (HELM Collaboration, 2023; BIG-bench Collaboration, 2023; Suzgun et al., 2023; Kwan et al., 2024). Human-preference platforms such as MT-Bench and Chatbot Arena also show that pairwise comparisons can be useful while still needing careful controls over position, sampling, and judge behavior (Zheng et al., 2023; Chiang et al., 2024). Our controlled panel is not a human-preference arena, but it borrows the idea that a pairwise comparison is the natural unit when the reader wants to decide between two systems.

Benchmark sources. The experiment uses established public benchmark families rather than a synthetic task suite: MMLU, ARC-Challenge, HellaSwag, GSM8K, and TruthfulQA (Hendrycks et al., 2021; Clark et al., 2018; Zellers et al., 2019; Cobbe et al., 2021; Lin et al., 2022). We also cite WinoGrande, BBQ, BIG-bench/BBH, and HELM as evaluation-context anchors rather than executed tasks (Sakaguchi et al., 2021; Parrish et al., 2022; BIG-bench Collaboration, 2023; Suzgun et al., 2023; HELM Collaboration, 2023). The style of broad benchmark reporting follows recent benchmark papers such as M4LE (Kwan et al., 2024), but our contribution is not a new dataset.

3 Method

Figure 1 summarizes the audit. Let a scored row contain a model, benchmark family, metric, evaluator-format axis, and official score. For each benchmark family and metric, we compare all unordered model pairs. A pair receives a stable decision only when the selected procedure’s interval excludes zero in one direction; otherwise it is abstained. The simultaneous-band procedure applies a Bonferroni-style correction across the family/metric pair set, so it is designed to reduce over-reporting of marginal wins.

The protocol has three safeguards. First, the

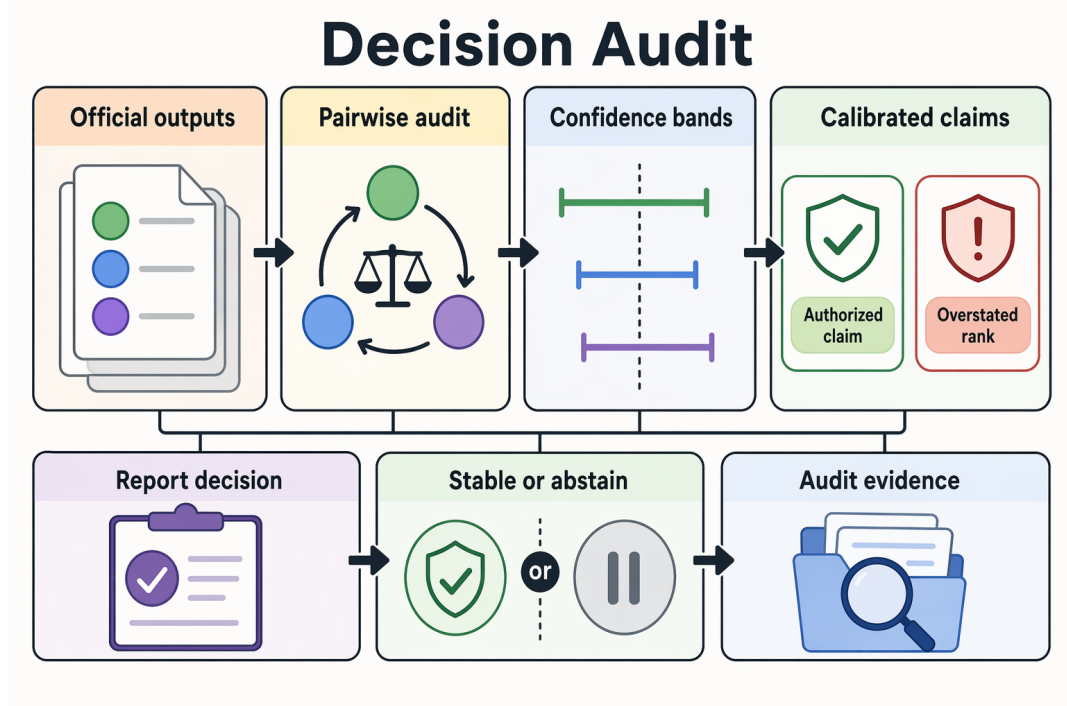


Figure 1: Decision-audit overview: official lm-eval outputs become pairwise stable-or-abstain decisions, with 15/315 stable simultaneous-band decisions in the controlled panel.

audit is run on a controlled model panel, so it is not allowed to infer public leaderboard retractions unless versioned leaderboard rows are captured. Second, the same-output baselines are named as style comparisons rather than exact reproductions of PromptEval or Rank Intervals. Third, unsupported axes remain explicit: shots, seeds, option-order permutations, and GSM8K extraction/filter variants are not part of the full tensor.

The decision labels are intentionally claim-level labels. A stable decision means that the paper is willing to report a direction for that model pair under the selected family, metric, and uncertainty procedure. An abstention means that the evidence should remain in the table but should not be converted into a pairwise model-selection recommendation. An axis flip means that the two observed evaluator-format settings disagree in direction; it is diagnostic evidence, not a simultaneous-inference decision. This separation prevents the draft from treating every sensitivity signal as a public error in a leaderboard.

Decision unit. For a benchmark family b , metric m , and model pair (i, j) , the audit computes a score difference Δ_{ijbm} from the saved scorer tensor. The same pair may appear under a naive interval, a simultaneous interval, and the observed

evaluator-format audit. The paper reports the stable/abstain label only after choosing the procedure. This avoids a common ambiguity in leaderboard discussion: a table may contain a numerical difference, a rank difference, and an uncertainty interval, but the prose may still state a win without naming which rule authorizes that claim.

Simultaneous bands. The simultaneous-band row uses the family/metric pair set as the multiple-comparison family. In this tensor, each family/metric cell contains 45 model pairs, so the correction is applied at the same granularity as the reader-facing claim. We use a Bonferroni normal approximation over that pairwise comparison family; the manuscript treats it as a conservative reporting rule, not as the only possible simultaneous-inference method. This is why Table 1 places naive and simultaneous rates next to each other: the comparison shows the cost of controlling the family of pairwise claims.

Axis audit. The evaluator-format audit compares the two observed format settings, chat-template and no-chat. A direction flip is counted when the model-pair ordering differs across those settings for the same family and metric. It is tempting to call every flip a false leaderboard decision, but the paper does not do that. The flip is evidence that a re-

portable claim should name the evaluation format, and it becomes especially relevant when simultaneous bands abstain because the margin is small. The flip count therefore helps explain why abstention is a useful output, while the simultaneous rule determines which claims are reportable.

Claim downgrade ledger. The last method component is a negative one: each planned claim class is checked against the evidence that would be needed to support it. Public leaderboard retractions require frozen leaderboard rows and model-version alignment. tinyBenchmarks claims require an IRT-calibrated subset validation rather than a smaller unvalidated sample. Full-grid claims require executed shot, seed, option-order, and extraction axes. The ledger is part of the method because it prevents the analysis from silently upgrading a controlled-panel result into a broader public claim.

4 Experimental Setup

The full run contains 2560 scored rows from the official lm-evaluation-harness scorer. The evaluated families are ARC-Challenge, GSM8K, HellaSwag, MMLU, and TruthfulQA; ARC-Challenge and HellaSwag each contribute two metrics, while GSM8K, MMLU, and TruthfulQA contribute one in the main pairwise audit. The observed evaluator-format axis has two levels, chat-template and no-chat.

The ten evaluated model identifiers are Qwen2.5-Instruct at 0.5B, 1.5B, 3B, 7B, 14B, and 32B; Mistral-7B-Instruct-v0.3; Mixtral-8x7B-Instruct-v0.1; Phi-3.5-mini-instruct; and OLMo-2-1124-13B-Instruct. The paper-facing budget is the crossed scorer tensor: 32 public examples per non-GSM8K family, 64 GSM8K examples, ten models, and two evaluator-format settings, with official harness scoring rather than a custom decoder or learned scorer.

The same-output baseline suite compares five analyses over the saved outputs: PromptEval-style marginalization summaries, rank-interval-style uncertainty summaries, benchmark-perturbation summaries, naive pair intervals, and simultaneous pair intervals. No IRT-calibrated tinyBenchmarks subset validation was executed, so all paper-facing subset claims remain out of scope. The supported claims use the controlled full tensor and same-output comparisons.

The benchmark rows are public task-family samples scored through the same harness interface. ARC-Challenge and HellaSwag each expose an

accuracy and normalized-accuracy view, GSM8K uses exact match, and MMLU and TruthfulQA use accuracy in the main audit. The paper does not combine these metrics into a single overall score because the reporting unit is a family/metric/model-pair claim. This keeps the statistical decision aligned with what a leaderboard user would ask: whether one model can be reported as better than another for a named benchmark family.

Executed tensor. The run matrix crosses ten model identifiers with two evaluator-format settings and five benchmark families. The task counts in the main results matrix are intentionally small but real public benchmark subsets: 32 examples for ARC-Challenge, HellaSwag, MMLU, and TruthfulQA, and 64 for GSM8K. This choice makes the paper a controlled-panel audit rather than a claim about exact public leaderboard placements. The same model/family/axis structure is nevertheless sufficient for the pairwise decision question, because every model pair is observed under the same family-level scoring contract.

Scoring and metrics. Multiple-choice tasks use the official harness scoring for the selected task configurations, while GSM8K is reported through exact match. The paper does not introduce a custom classifier, bag-of-words scorer, or hand-written oracle. It also does not use chain-of-thought self-consistency as a paper-facing method, even though self-consistency is a relevant GSM8K and reasoning-evaluation anchor (Wang et al., 2023). The evaluated object is the score tensor and the reporting protocol applied to it.

Baselines and controls. The same-output baselines are diagnostic controls over the saved tensor. PromptEval-style rows ask what happens when available prompt/output views are marginalized; rank-interval-style rows ask how uncertainty changes the rank or pairwise claim; benchmark-perturbation rows summarize family/metric pair flip rates. These baselines are not presented as exact reproductions of prior systems because the full prompt grid and public leaderboard snapshots are not present. They are used to make the decision audit interpretable against the closest prior designs while keeping the executed evidence honest.

Stopping and freshness. A run condition is included in the paper only when it has completed raw scored rows and is represented in the analysis summaries. The full tensor contains 2560 rows, and the

full baseline tensor contains 778 rows. Failed or blocked claim classes remain in the failure taxonomy and in the paper’s scoped claims. This stopping rule is conservative: it makes the paper less expansive, but it prevents planned axes or benchmark-construction files from being counted as executed evidence.

5 Results

Table 1 gives the central paper-facing result. Simultaneous bands report 15 stable decisions and 300 abstentions over 315 pairwise comparisons, a stable-decision rate of 4.8%.

Naive intervals report 43 stable decisions over the same 315 comparisons, and the axis audit reports 178 stable decisions with 137 abstentions before simultaneous correction.

The rows use the public lm-eval task-family subset for each benchmark (32 examples except 64 for GSM8K), the same ten-model open panel, and official scorer outputs under two evaluator-format settings. The table includes those replay fields next to the decision outcomes so the main comparison can be read without stitching together setup prose and appendix inventory rows.

The four result columns make Table 1 the paper-facing decision ledger rather than a run inventory: simultaneous stable decisions define reportable pairwise claims, naive stable decisions show the cost of a weaker interval rule, axis flips expose evaluator-format sensitivity, and simultaneous abstentions record withheld claims.

Table 1 is a claim ledger rather than a rank table. Its first contrast is the simultaneous decision rule: only 15/315 pairwise directions are stable enough to report, while 300/315 are withheld as abstentions. Its second contrast is the interval baseline, where a weaker naive rule would authorize 43/315 directions. Its third contrast is evaluator-format sensitivity: 53 pairs change direction across the two observed format settings, so the table separates diagnostic flips from reportable simultaneous-band claims. Figure 2 visualizes the same decision-rate contrast and makes the abstention-heavy outcome visible before the family-level analysis.

The strongest interpretation is conservative. The evidence does not show that public leaderboards should be rewritten; it shows that, in this controlled panel, many pairwise claims lack enough support under a simultaneous reporting rule. This is the decision-theoretic value of the audit: abstention is

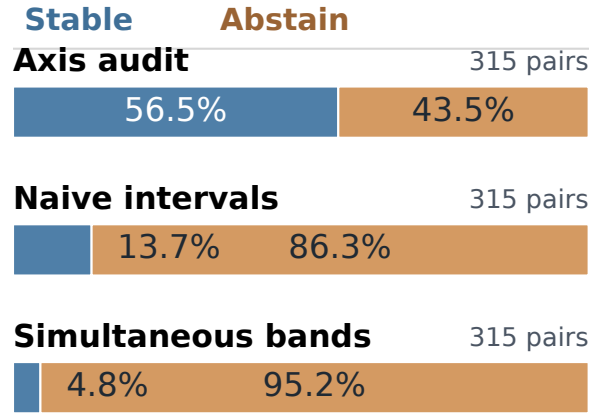


Figure 2: Decision stability rates from the analysis tables: simultaneous bands reduce the stable-decision rate to 4.8% and make abstention the dominant reporting outcome.

a measured outcome rather than a failure to produce a rank.

Family-level contrast. The strongest contrasts are family-specific. MMLU and HellaSwag accuracy have zero simultaneous stable decisions in this tensor, while GSM8K exact match retains six stable decisions.

The raw axis audit is less conservative, but it is not a reporting protocol by itself because it does not correct across all pairwise claims. The gap between the raw axis audit, naive intervals, and simultaneous bands is therefore the central empirical result: a leaderboard can contain many visible differences while still supporting only a small number of reportable pairwise conclusions under a simultaneous standard.

Table 1 also shows why a single aggregate rank would be misleading for this study. ARC-Challenge normalized accuracy has four simultaneous stable decisions, while ARC-Challenge raw accuracy has one; HellaSwag normalized accuracy has four, while HellaSwag raw accuracy has none.

If those family/metric cells were averaged into a single score, the reader would lose the fact that support for a pairwise conclusion depends on both benchmark family and metric. The paper therefore keeps the family/metric cells separate even though this makes the table more detailed.

Naive intervals. The naive interval baseline is useful precisely because it is easier to satisfy. It reports 43 stable decisions, almost three times the simultaneous count. A paper that only reported

Benchmark / metric	Source / sample	Panel / backend	Budget / metric	Simultaneous band	Naive	Format flips
ARC-Challenge / acc.	lm-eval public; 32 ex.	10 open models; official scorer	2 formats; acc.	1/45 stable; 44/45 withheld	7/45 (15.6%)	1/45
ARC-Challenge / norm. acc.	lm-eval public; 32 ex.	10 open models; official scorer	2 formats; norm. acc.	4/45 stable; 41/45 withheld	11/45 (24.4%)	3/45
GSM8K / exact match	lm-eval public; 64 ex.	10 open models; official scorer	2 formats; exact match	6/45 stable; 39/45 withheld	17/45 (37.8%)	8/45
HellaSwag / acc.	lm-eval public; 32 ex.	10 open models; official scorer	2 formats; acc.	0/45 stable; 45/45 withheld	0/45 (0.0%)	4/45
HellaSwag / norm. acc.	lm-eval public; 32 ex.	10 open models; official scorer	2 formats; norm. acc.	4/45 stable; 41/45 withheld	6/45 (13.3%)	8/45
MMLU / acc.	lm-eval public; 32 ex.	10 open models; official scorer	2 formats; acc.	0/45 stable; 45/45 withheld	2/45 (4.4%)	18/45
TruthfulQA / acc.	lm-eval public; 32 ex.	10 open models; official scorer	2 formats; acc.	0/45 stable; 45/45 withheld	0/45 (0.0%)	11/45

Table 1: Simultaneous bands authorize 15/315 decisions; naive intervals authorize 43/315, and format changes flip 53 pairs.

naive intervals could therefore make many more directional claims from the same tensor. The simultaneous protocol rejects that temptation: it trades report count for a cleaner family-wise reporting standard. This tradeoff is the practical recommendation of the paper, not a claim that Bonferroni correction is always optimal.

Format-axis flips. The axis audit provides a complementary warning. It reports 178 directional decisions before simultaneous correction, but 53 of the 315 pairs change direction between the two observed evaluator-format settings. Those flips are concentrated enough to matter for prose, yet sparse enough that aggregate variance shares remain small. This combination is the main qualitative lesson: score variance and decision instability are related but not interchangeable. Pairwise margins can be fragile even when the evaluator-format axis explains only a modest share of total score variation.

Table 1 therefore motivates the analysis below: the variance plot checks whether evaluator format dominates aggregate score variation, while the axis audit shows why small aggregate shares can still matter for pairwise claims.

6 Analysis and Ablations

The analysis separates two claims that Table 1 leaves adjacent: evaluator-format changes can flip some pairwise directions, but aggregate variance shares remain bounded in the observed tensor.

The evaluator-format axis still matters even when it explains a small share of score variance.

The axis audit records 53 direction-flipping pairs over 315 family/metric/model-pair comparisons.

The variance decomposition attributes between 0.0% and 1.9% of family/metric score variation to the observed design axis, with the largest observed design-axis share in ARC-Challenge normalized accuracy. These shares are not estimates for shot, seed, option-order, or extraction axes; those factors were not executed in the full tensor.

These shares matter for reporting because aggregate variance and pairwise claim stability answer different questions. A low average design-axis share can still move a small-margin model pair, while a large model-identifier share does not by itself authorize a directional claim.

The same-output baselines clarify what this protocol adds and what it does not. PromptEval-style rows summarize marginalization over available outputs, and rank-interval-style rows summarize uncertainty. The decision audit combines intervalized pairwise reporting with an explicit downgrade ledger, so its main output is a calibrated claim set. It is therefore an audit layer over existing scored outputs, not a new learned model, new benchmark, or exact reimplementations of each prior method.

Figure 3 compares the model-identifier share with the observed evaluator-format share for each family/metric cell. It is a guardrail against a different overclaim: the largest recorded design-axis share is 1.9%, and several family/metric cells are much smaller.

That does not make the axis irrelevant. Decision instability depends on pairwise margins as well as aggregate variance. It does mean that a

full variance-decomposition story over prompt templates, shots, seeds, option order, and extraction rules would require those axes to be run rather than inferred.

Why small variance can still flip claims. A family-level variance share averages over models and examples. A pairwise claim, by contrast, depends on a margin between two particular models. When the margin is small, even a low-variance axis can move the direction of the difference. This is the situation captured by the axis-flip rows: they are not evidence that the evaluator-format axis dominates the benchmark, but they are evidence that some rank-adjacent comparisons should not be reported without naming the format condition.

Why simultaneous abstention is not a failure. The simultaneous rule abstains on 300/315 comparisons. In a leaderboard culture this can look like information loss, because abstention removes many directional claims. In a claim-audit culture it is information: it tells the reader that the score tensor does not justify a pairwise recommendation under the selected family-wise rule. The distinction aligns with prior calls for reporting uncertainty and experimental conditions rather than only point estimates (Dodge et al., 2019; Card et al., 2020; Dror et al., 2018).

Relationship to prior robust-ranking work. PromptEval, rank intervals, and related graph-based perturbation methods are stronger than a naive one-rank leaderboard in different ways (Polo et al., 2024b; Neuhoof and Benjamini, 2026; Huang et al., 2026).

The current audit is not proposed as a replacement for those methods. Its role is a conservative reporting layer over a completed tensor: when a prior method or baseline suggests uncertainty, the audit translates that uncertainty into a paper-facing stable-or-abstain label and records which claims cannot be made.

What would change the conclusion. The conclusion would become stronger if a future run added the omitted grid axes and still found the same abstention pattern. It would also become more public-facing if exact Open LLM Leaderboard rows were frozen and model versions were matched. Conversely, if a larger task sample made many more intervals exclude zero, the simultaneous stable-decision rate would rise and the paper’s conservative recommendation would weaken. The

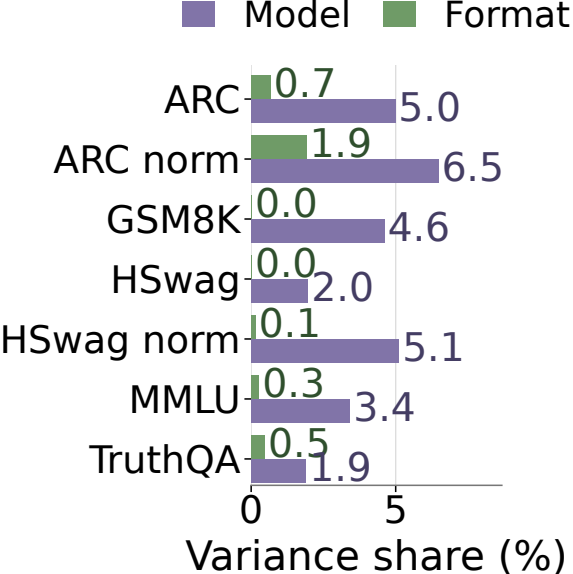


Figure 3: Observed design-axis variance shares stay at or below 1.9% in the completed tensor; unexecuted original-grid axes remain unsupported.

current result is therefore a property of the executed tensor and protocol, not a universal law about all leaderboards.

7 Failure Cases

The failure ledger contains three hard boundaries. First, public leaderboard retraction claims are blocked because exact versioned leaderboard rows were not captured. Second, tinyBenchmarks/IRT subset validity is blocked because the full tensor uses deterministic official samples rather than an IRT-calibrated subset. Third, the original grid axes beyond chat-template/no-chat were not run. These gaps are not hidden in limitations; they determine the wording of the main claims.

The paper consequently avoids false-win language. Without external ground-truth labels for model superiority, an unstable or abstained pair is not a false win. It is a reporting decision that the current evidence does not certify.

These failure cases are useful because they define the minimum evidence needed for a stronger future paper. Public retraction claims need frozen leaderboard rows and model/version matching. Subset-efficiency claims need an IRT-calibrated sampling validation, not merely a smaller task file. Full-grid claims need the missing axes to be executed with the same scorer and model panel. Until then, the conservative statement is the controlled-panel one reported here.

Public leaderboard retractions. The broader research direction asked whether published pairwise leaderboard conclusions could be overturned after marginalizing over a full defensible choice grid. That is an attractive claim, but it is not supported by the current evidence. The controlled panel identifies abstentions; it does not say that a named public leaderboard comparison should be retracted.

tinyBenchmarks and IRT. tinyBenchmarks is relevant because it motivates compute-efficient evaluation (Polo et al., 2024a), but the completed tensor does not validate an IRT-calibrated subset. The completed baseline analysis does not contain a calibrated subset validation. The manuscript therefore cites tinyBenchmarks as prior work and records the missing validation as a limitation. It does not use the phrase “tinyBenchmarks result” for the completed run.

Unexecuted design axes. Extraction rules apply differently across benchmark families. Multiple-choice tasks are scored with loglikelihood-style official scoring, while GSM8K exact match is the relevant generative/extraction case. Because the full tensor did not execute shot, seed, option-order, and extraction-rule grids, the paper cannot present a uniform five-benchmark full-grid variance decomposition. The correct claim is narrower: the observed chat-template/no-chat axis is enough to motivate a decision audit, but it is not enough to characterize every defensible evaluator choice.

Reporting template implied by the failures. The failure taxonomy leads to a concrete reporting template. A benchmark report should name the benchmark family and metric, the model pair, the scorer or evaluator-format family, the uncertainty procedure, the stable-or-abstain label, and the evidence class that would be needed for any stronger claim. For this tensor, a valid report can say that GSM8K exact match retains 6 stable simultaneous-band decisions out of 45 model pairs. It should not say that the same finding proves public leaderboard retractions, because the public rows are not frozen. Likewise, it can say that HellaSwag accuracy has 0 stable simultaneous-band decisions in the controlled tensor; it should not say that all HellaSwag leaderboard differences are meaningless.

How to read an abstention. An abstention is neither a tie nor a model-quality judgment. It says that the chosen evidence package does not support a directional pairwise statement. This distinction

matters because the same pair can have a nonzero point difference, a naive interval that excludes zero, and a simultaneous interval that includes zero. The audit asks authors to state which level of evidence they are using. If they want the stronger public claim, the failure ledger says what must be added: versioned public rows, exact model matching, more evaluator axes, or a validated subset design.

Why the controlled panel is still useful. The controlled panel is not a substitute for a full leaderboard archive, but it has one advantage: every model pair is observed under the same task-family and evaluator-format contract. That makes the pairwise decision labels internally comparable. The panel therefore functions as a rehearsal for a larger public audit. If the same procedure were applied to frozen public leaderboard rows, the output would be immediately interpretable: stable public claims, abstained public claims, and claims that require more evidence because the published records do not expose enough evaluator-choice information.

Minimum stronger-evidence package. The next stronger evidence package would add three components. First, it would freeze a public leaderboard snapshot with model identifiers, timestamps, task versions, and official aggregation rules. Second, it would execute the missing defensible axes that apply to each task family, without pretending that GSM8K extraction and multiple-choice loglikelihood scoring are the same design dimension. Third, it would validate any compute-saving subset against the target task distribution. Until those three components exist, the current paper should remain a controlled-panel abstention study.

8 Conclusion

This controlled-panel study converts leaderboard rankings into audited pairwise reporting decisions. On the completed tensor, simultaneous bands make abstention the dominant outcome, with 15 stable decisions and 300 abstentions over 315 comparisons. The result is a practical reporting standard: state stable pairwise claims when the audit supports them, and otherwise report abstention and the missing evidence.

9 Limitations

The study is bounded to ten open instruction/chat models, five benchmark families, two observed evaluator-format axes, and same-output baselines.

It does not provide public leaderboard retractions, tinyBenchmarks/IRT validation, or full-grid estimates for shots, seeds, option order, and GSM8K extraction/filter choices. The intervals use a normal-approximation implementation over the saved tensor and should be treated as a reporting audit, not as a universal uncertainty model for all leaderboards.

The benchmark samples are public and official, but they are controlled subsets rather than the exact public leaderboard submissions. This matters because leaderboard rows often combine task versions, model revisions, prompt conventions, and aggregation rules that are not recoverable from a rank table alone. A public retraction claim would require frozen rows, model-version alignment, and the same scoring contract used by the published table. The current evidence deliberately stops short of that claim.

The omitted evaluator choices also matter. Multiple-choice loglikelihood scoring and GSM8K exact-match extraction are not the same design axis, and a future full-grid audit would need task-specific axes rather than a uniform extraction column across all families. Similarly, an IRT-calibrated subset claim would require validating the selected subset against the target task distribution. The current controlled panel is therefore useful for a reporting protocol, but it should not be read as a final statement about every defensible evaluation choice.

The study also does not train or adapt a model. That is appropriate for the narrowed thesis because the method is a reporting protocol over existing scorer outputs, but it means the paper should not be read as a new modeling result. Finally, the same-output baselines are comparison anchors, not exact prior-work reproductions under every published design choice.

10 Ethical Considerations

The work uses public benchmark families and open model identifiers. Its main ethical effect is cautionary: it discourages overconfident claims from narrow leaderboard differences and unfrozen public rows. A stable-or-abstain report protects readers from treating a format-sensitive gap as settled and protects developers from unsupported public retraction claims. Abstention should be described as an evidence-and-uncertainty statement, not as a tie, failure, or model-quality judgment.

References

- Norah Alzahrani, Hisham Alyahya, Yazeed Alnumay, Sultan AlRashed, Shaykhah Alsubaie, Yousef Almushayqih, Faisal Mirza, Nouf Alotaibi, Nora Al-Twairish, Areeb Alowisheq, M Saiful Bari, and Haidar Khan. 2024a. [When benchmarks are targets: Revealing the sensitivity of large language model leaderboards](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*.
- Norah Alzahrani, Hisham Abdullah Alyahya, Yazeed Alnumay, Sultan AlRashed, Shaykhah Alsubaie, Yousef Almushaykeh, Faisal Mirza, Nouf Alotaibi, Nora Altwairish, and Areeb Alowisheq. 2024b. [When benchmarks are targets: Revealing the sensitivity of large language model leaderboards](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*.
- Angel Rodrigo Avelar Menendez, Yufeng Liu, and Xiaowu Dai. 2026. Prompt-dependent ranking of large language models with uncertainty quantification. *arXiv preprint*.
- BIG-bench Collaboration. 2023. Beyond the imitation game: Quantifying and extrapolating the capabilities of language models. *Transactions on Machine Learning Research*.
- Dallas Card, Peter Henderson, Urvashi Khandelwal, Robin Jia, Kyle Mahowald, and Dan Jurafsky. 2020. [With little power comes great responsibility](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*.
- Wei-Lin Chiang, Lianmin Zheng, Ying Sheng, Anastasios N. Angelopoulos, Tianle Li, Dacheng Li, Hao Zhang, Michael I. Jordan, Joseph E. Gonzalez, and Ion Stoica. 2024. Chatbot arena: An open platform for evaluating llms by human preference. In *Proceedings of the 41st International Conference on Machine Learning*.
- Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. 2018. Think you have solved question answering? try arc, the ai2 reasoning challenge. *arXiv preprint*.
- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, and Reiichiro Nakano. 2021. Training verifiers to solve math word problems. *arXiv preprint*.
- Contrast Sets Authors. 2020. [Evaluating models' local decision boundaries via contrast sets](#). In *Findings of the Association for Computational Linguistics: EMNLP 2020*.
- Jesse Dodge, Suchin Gururangan, Dallas Card, Roy Schwartz, and Noah A. Smith. 2019. [Show your work: Improved reporting of experimental results](#). In *Proceedings of the 2019 Conference on Empirical*

767	<i>Methods in Natural Language Processing and the 9th</i>	Stephanie Lin, Jacob Hilton, and Owain Evans. 2022.	820
768	<i>International Joint Conference on Natural Language</i>	Truthfulqa: Measuring how models mimic human	821
769	<i>Processing.</i>	falsehoods. In <i>Proceedings of the 60th Annual Meeting</i>	822
		<i>of the Association for Computational Linguistics.</i>	823
770	Rotem Dror, Gili Baumer, Segev Shlomov, and Roi	LM Evaluation Harness Contributors. 2023. A	824
771	Reichart. 2018. The hitchhiker’s guide to testing	framework for few-shot language model evaluation.	825
772	statistical significance in natural language processing.		
773	In <i>Proceedings of the 56th Annual Meeting of the</i>	Yanan Long. 2026. Bayesian inference and decision	826
774	<i>Association for Computational Linguistics (Volume 1:</i>	audits for public archives of frontier ai evaluations.	827
775	<i>Long Papers).</i>	<i>arXiv preprint.</i>	828
776	Kawin Ethayarajh and Dan Jurafsky. 2020. Utility is in	Bitya Neuhof and Yuval Benjamini. 2026. Rank	829
777	the eye of the user: A critique of NLP leaderboards. In	intervals for leaderboards: A hierarchical framework	830
778	<i>Proceedings of the 2020 Conference on Empirical</i>	for model evaluation. <i>arXiv preprint.</i>	831
779	<i>Methods in Natural Language Processing.</i>		
780	David Heineman, Valentin Hofmann, Ian Magnusson,	Hosna Oyarhoseini, Jimmy Lin, and Amir-Hossein	832
781	Yuling Gu, Noah A. Smith, Hannaneh Hajishirzi, Kyle	Karimi. 2026. A unified perturbation framework for	833
782	Lo, and Jesse Dodge. 2025. Signal and noise: A	analyzing leaderboard stability and manipulation.	834
783	framework for reducing uncertainty in language model	<i>arXiv preprint.</i>	835
784	evaluation. <i>arXiv preprint.</i>		
785	HELM Collaboration. 2023. Holistic evaluation of	Kishore Papineni, Salim Roukos, Todd Ward, and	836
786	language models. <i>Transactions on Machine Learning</i>	Wei-Jing Zhu. 2002. BLEU: a method for automatic	837
787	<i>Research.</i>	evaluation of machine translation. In <i>Proceedings of</i>	838
		<i>the 40th Annual Meeting of the Association for</i>	839
		<i>Computational Linguistics.</i>	840
788	Dan Hendrycks, Collin Burns, Steven Basart, Andy	Alicia Parrish, Angelica Chen, Nikita Nangia, Vishakh	841
789	Zou, Mantas Mazeika, Dawn Song, and Jacob	Padmakumar, Jason Phang, Jana Thompson, Phu Mon	842
790	Steinhardt. 2021. Measuring massive multitask	Htut, and Samuel R. Bowman. 2022. BBQ: A	843
791	language understanding. In <i>International Conference</i>	hand-built bias benchmark for question answering. In	844
792	<i>on Learning Representations.</i>	<i>Findings of the Association for Computational</i>	845
		<i>Linguistics: ACL 2022.</i>	846
793	Dong Huang, Jianbo Sun, and Pengkun Yang. 2026.	Felipe Maia Polo, Lucas Weber, Leshem Choshen,	847
794	Prompt perturbation for reliable llm evaluation over	Yuekai Sun, Gongjun Xu, and Mikhail Yurochkin.	848
795	comparison graphs. <i>arXiv preprint.</i>	2024a. tinybenchmarks: Evaluating llms with fewer	849
		examples. In <i>Proceedings of the 41st International</i>	850
796	Bora Kargi and David Salinas. 2026. From uncertain	<i>Conference on Machine Learning.</i>	851
797	judgments to calibrated rankings: Conformal elo		
798	estimation for llm evaluation. <i>arXiv preprint.</i>	Felipe Maia Polo, Ronald Xu, Lucas Weber, Mirian	852
799	Douwe Kiela, Max Bartolo, Yixin Nie, Divyansh	Silva, Onkar Bhardwaj, Leshem Choshen, Allysson	853
800	Kaushik, Atticus Geiger, Zhengxuan Wu, Bertie	Flavio Melo de Oliveira, Yuekai Sun, and Mikhail	854
801	Vidgen, Grusha Prasad, Amanpreet Singh, Pratik	Yurochkin. 2024b. Efficient multi-prompt evaluation of	855
802	Ringshia, Zhiyi Ma, Tristan Thrush, Sebastian Riedel,	llms. In <i>Advances in Neural Information Processing</i>	856
803	Zeeraq Waseem, Pontus Stenetorp, Robin Jia, Mohit	<i>Systems.</i>	857
804	Bansal, Christopher Potts, and Adina Williams. 2021.	Amirhossein Razavi, Mina Soltangheis, Negar	858
805	Dynabench: Rethinking benchmarking in NLP. In	Arabzadeh, Sara Salamat, Morteza Zihayat, and	859
806	<i>Proceedings of the 2021 Conference of the North</i>	Ebrahim Bagheri. 2025. Benchmarking prompt	860
807	<i>American Chapter of the Association for Computational</i>	sensitivity in large language models. <i>arXiv preprint.</i>	861
808	<i>Linguistics: Human Language Technologies.</i>		
809	Wai-Chung Kwan, Xingshan Zeng, Yufei Wang, Yusen	Keisuke Sakaguchi, Ronan Le Bras, Chandra	862
810	Sun, Liangyou Li, Yuxin Jiang, Lifeng Shang, Qun Liu,	Bhagavatula, and Yejin Choi. 2021. Winogrande: An	863
811	and Kam-Fai Wong. 2024. M4LE: A multi-ability	adversarial winograd schema challenge at scale. In	864
812	multi-range multi-task multi-domain long-context	<i>Proceedings of the AAAI Conference on Artificial</i>	865
813	evaluation benchmark for large language models. In	<i>Intelligence.</i>	866
814	<i>Proceedings of the 62nd Annual Meeting of the</i>	Mirac Suzgun, Nathan Scales, Nathanael Schärli,	867
815	<i>Association for Computational Linguistics (Volume 1:</i>	Sebastian Gehrmann, Yi Tay, Hyung Won Chung,	868
816	<i>Long Papers).</i>	Aakanksha Chowdhery, Quoc V. Le, Ed H. Chi, Denny	869
		Zhou, and Jason Wei. 2023. Challenging big-bench	870
817	Chin-Yew Lin. 2004. ROUGE: A package for	tasks and whether chain-of-thought can solve them. In	871
818	automatic evaluation of summaries. In <i>Text</i>	<i>Findings of the Association for Computational</i>	872
819	<i>Summarization Branches Out.</i>	<i>Linguistics: ACL 2023.</i>	873

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations*.

Rowan Zellers, Ari Holtzman, Yonatan Bisk, Ali Farhadi, and Yejin Choi. 2019. Hellaswag: Can a machine really finish your sentence? In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging llm-as-a-judge with mt-bench and chatbot arena. In *Advances in Neural Information Processing Systems*.

A Reproducibility Appendix

The replay package contains the scored official-harness tensor, same-output baselines, analysis tables, and data-figure sources: 2560 scorer rows and 778 baseline rows. The paper-facing boundary is evidence classes and replay steps, not implementation paths. Tables 2 and 3 identify the public samples and the missing evidence for stronger claims.

Run inventory. Table 2 records the public sample and metric for each family. All rows use the ten-model panel, the official lm-evaluation-harness scorer, and the two observed evaluator-format settings.

Benchmark	Sample	Metrics
ARC-Challenge	32 ex.	acc.; norm. acc.
GSM8K	64 ex.	exact match
HellaSwag	32 ex.	acc.; norm. acc.
MMLU	32 ex.	acc.
TruthfulQA	32 ex.	acc.

Table 2: Five public benchmark samples and 778 same-output baseline rows define the replay panel.

Claim downgrade ledger. Table 3 lists the three scope limits from Section 7. Each row names the evidence still missing before a stronger claim is available. Until that evidence is executed or archived, the weaker controlled-panel claim is the paper’s claim.

Claim class	Evidence still missing
Public rows	Frozen leaderboard rows with model versions
Subset validity	Calibrated tinyBenchmarks/IRT check
Full-grid axes	Executed shot, seed, option-order, and extraction rows

Table 3: Three claim classes remain scoped until missing evidence is added.

Allowed current claims are narrow: public-row questions support Controlled-panel abstentions only; subset questions motivate prior work; full-grid questions support only the two-format axis audit.

Replay interface. The paper-facing replay interface has three steps. First, rerun the official scorer matrix for the same public subsets and ten model identifiers. Second, rebuild the same-output baseline tensor, keeping PromptEval-style, rank-interval-style, benchmark-perturbation, naive-pair,

and simultaneous-pair rows distinct. Third, regenerate the manuscript tables and data figures from the analysis tables.

Seed and sampling policy. The saved public subsets and scorer outputs make the completed tensor deterministic, so the paper does not report a multi-seed generation study. Future stochastic generation, prompt sampling, or option-order permutations should appear as executed variance factors rather than as part of the present evaluator-format axis.

Evidence package inventory. The supplement should contain public benchmark provenance, model-panel metadata, raw scored rows, same-output baseline rows, analysis tables, data figures, manuscript source, and the compiled PDF. Low-level provenance can remain supplementary; the manuscript reports benchmarks, model identifiers, metrics, task counts, and scoring policy.

Artifact map. The release bundle maps each result to one artifact family: decision stability summaries for Table 1, variance rows for Figure 3, failure-taxonomy rows for Table 3, and paired-significance rows for simultaneous-band decisions. This map separates simultaneous-band decisions from the raw evaluator-format audit.

Verification and artifact map. The main numerical claims reduce to four replay checks: 2560 full scored rows, 778 same-output baseline rows, 15/315 simultaneous stable decisions with 300 abstentions, and 53 evaluator-format direction flips. The decision-stability and paired-significance rows back the abstention claim; the variance table and Figure 3 back the variance-boundary claim; the failure taxonomy backs the unsupported-grid claim; and the absence of frozen public rows backs the public-leaderboard downgrade.

A reader can reproduce Table 1 by grouping the decision-stability summary by procedure, benchmark family, and metric, then summing total pairs, stable decisions, and abstentions for simultaneous, naive, and axis-audit rows. Reproducing Figure 3 requires grouping the variance table by family, metric, and factor and plotting the design-axis and model-identifier shares. A regenerated paper that changes any of these values must update the abstract, Results section, captions, and claim statements together.

Failure and packaging boundary. Three stronger claims should remain absent unless new evidence is added. Exact public leaderboard overturning requires frozen public rows and model-version mappings. tinyBenchmarks claims require calibrated IRT validation. Shot, seed, option-order, or extraction-rule variance claims require executed rows for those factors. The submission package should include paper source, bibliography, generated figures, result tables, and replay metadata for the public benchmark tensor, but not model weights, private cache directories, host-specific execution paths, or lower-level provenance better kept in supplementary metadata.